4COSC002W Mathematics for Computing

Lecture 8

Matrices – Part 2.

Power of a Matrix. Inverse of a Matrix

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Power of a Matrix

The *power of a matrix* is a concept similar to the power of numbers, but it applies to matrices.

When we say a matrix is raised to a power, it means the matrix is multiplied by itself a certain number of times.

It's important to note that matrix multiplication is not commutative (i.e., $AB \neq BA$ in general), so the power of matrices applies to square matrices only.

For a **square matrix A** (a matrix with the same number of rows and columns), **the power of the matrix is denoted as $\mathbf{A^n}$**, where n is a positive integer. It is defined as:

$$A^1 = A$$

$$A^2 = A \times A$$

$$A^3 = A \times A \times A$$

And so on, where A^n involves multiplying A by itself n times.

Example 1 of Power of a Matrix

Suppose we have the following **square matrix *A***:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Now, let's calculate $\mathbf{A^2}$, which means multiplying matrix A by itself: $A^2 = A \times A$

We can perform this multiplication to find A^2:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

To calculate the product, we multiply the rows of the first matrix by the columns of the second matrix:

$$A^2 = \begin{bmatrix} (2 \times 2 + 1 \times 1) & (2 \times 1 + 1 \times 3) \\ (1 \times 2 + 3 \times 1) & (1 \times 1 + 3 \times 3) \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 10 \end{bmatrix}$$

Example 2 of Power of a Matrix

Let's use a 3x3 **diagonal matrix *D*** with diagonal elements of 3:

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Now, let's calculate $\boldsymbol{D^3}$:

$$D^3 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3^3 & 0 & 0 \\ 0 & 3^3 & 0 \\ 0 & 0 & 3^3 \end{bmatrix} = \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}$$

The Inverse of a Matrix

For an invertible **matrix** $\boldsymbol{A}$, its **inverse** $\boldsymbol{A}^{-1}$ is a matrix that *"undoes"* the effect of A in matrix multiplication.

When A is multiplied by its inverse A^{-1}, the result is the identity matrix I, which acts like the number 1 in matrix operations:

$$A \times A^{-1} = I$$

Why Use the Inverse?

In regular arithmetic, we "undo" multiplication with division. For matrices, there is no division operation, so we use the inverse instead.

To solve equations like

$$A \times X = B,$$

we "divide" by A by multiplying by its inverse:

$$X = A^{-1} \times B$$

This isolates X, similar to how division works in regular algebra.

Key Point: The inverse of a matrix is essential in applications where we need to "reverse" or "undo" matrix transformations, like solving systems of equations or decoding encrypted messages.

Non-Singular Matrices

Not all matrices have inverses.

A matrix that has an inverse is called an *invertible or nonsingular matrix*.

If a matrix doesn't have an inverse, it's called a singular matrix.

The key requirement for a matrix to be invertible:

- square (having the same number of rows and columns)
- determinant must be non-zero.

The Determinant of a Square Matrix

The *determinant of a square matrix* is a scalar value that can be computed from the elements of the matrix. It is a fundamental concept in linear algebra and is denoted by "det(A)" or "|A|," where "A" is the matrix in question.

The determinant of a square matrix, such as an $n \times n$ matrix, is defined as follows:

- For a 1x1 matrix (a single number), the determinant is the number itself.
- For a 2x2 matrix:

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

For larger square matrices ($n \times n$ where $n > 2$), the determinant can be computed using various methods, such as the expansion by minors, cofactor expansion, or using specialised algorithms like Gaussian elimination.

Inverse of a 2x2 Matrix

A 2x2 matrix A is invertible if its determinant (det(A)) is non-zero.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{det(A) = ad - bc}$$

Inverse Calculation:

If det(A) ≠ 0, the inverse of A is:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The inverse is obtained by multiplying the reciprocal of the determinant by the adjugate of A.

Example: Inverse of a 2x2 Matrix

Let A be a matrix where:

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

First, we find the determinant of A: $\det(A) = (4)(6) - (7)(2) = 24 - 14 = 10$

Since $\det(A) \neq 0$, matrix A is invertible. Now, we'll use the formula for the inverse of a 2x2 matrix:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

When multiplied by the original matrix A, this inverse matrix will yield the identity matrix, confirming that it is indeed the correct inverse.

Determinant of a 3x3 Matrix: Diagonal Method

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right]$$

Determinant (|A|):

The sum of the products of the down diagonals minus the sum of the products of the up diagonals.

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ -(a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33})$$

Example:
Determinant of a 3x3 Matrix

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

To use the diagonal method, we need to visualise the matrix with two additional columns that repeat the first two columns of the matrix to the right.

$$\begin{array}{ccccc} 1 & 2 & 3 & 1 & 2 \\ 4 & 5 & 6 & 4 & 5 \\ 7 & 8 & 9 & 7 & 8 \end{array}$$

$$\text{Det(B)} = (1 \times 5 \times 9) + (2 \times 6 \times 7) + (3 \times 4 \times 8) - (3 \times 5 \times 7) - (1 \times 6 \times 8) - (2 \times 4 \times 9) = 0$$

So, matrix B is a *singular matrix*.

Determinant of a 3x3 Matrix: Minors and Cofactors

Square Matrix A:

Elements are denoted by a_{ij} where i and j represent the row and column indices.

Minors (Mij): The minor of element a_{ij} is the determinant of the 2x2 matrix formed by removing the i^{th} row and j^{th} column from A.

Cofactors (Cij): The cofactor of a_{ij} is calculated as

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

This considers the position of the element within the matrix by alternating signs.

Determinant Calculation: To find the determinant of A, multiply each element by its cofactor and sum the products for any row or column.

Determinant of a 3x3 Matrix: Minors Calculation

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Element	Minor	Element	Minor
a_{11}	$M_{11} = \det\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$	a_{22}	$M_{22} = \det\begin{bmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{bmatrix}$
a_{21}	$M_{21} = \det\begin{bmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{bmatrix}$	a_{23}	$M_{23} = \det\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}$
a_{31}	$M_{31} = \det\begin{bmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{bmatrix}$	a_{33}	$M_{33} = \det\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Minors Calculation Formulae for 3 by 3 matrix

Determinant of a 3x3 Matrix: Cofactors Calculation

$$\begin{vmatrix} + & - \\ - & + \end{vmatrix} \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} \quad \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$$

$$\left|C_{ij}\right| \equiv (-1)^{i+j}\left|M_{ij}\right|$$

The pattern of signs for cofactor minors

Determinant of a 3x3 Matrix: Minors and Cofactors

Cofactor Definition: For element a_{ij} in an $n \times n$ matrix, the cofactor C_{ij} is found using $(-1)^{i+j}M_{ij}$, where M_{ij} is the minor of a_{ij}.

Determinant of 3x3 Matrix:

1. Pick ANY *row* or *column*.

2. Calculate the *minors*. Multiply the *minor of each element* by $+1$ or -1 based on the position (sum of indices even or odd) to get the *cofactors*.

3. Multiply each element by its *cofactor* and sum up these products to get the determinant.

Formula Example (for the first row):

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

Example:
Finding the determinant using Minors and Cofactors Method

Given the matrix A:

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{pmatrix}$$

Task:

- Find the minors and cofactors for all elements in the second row.
- Use the second row to calculate the determinant of A by cofactor expansion.

Example:
Finding the determinant using Minors and Cofactors Method

Step 1: Minors and Cofactors for the *Second* Row

Element $a_{21} = 4$; **Minor** $\mathbf{M_{21}}$: Remove the second row and first column: $\begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix} = (0 \times 1) - (3 \times -2) = 6$

Cofactor C_{21}: Since $(2 + 1)$ is odd, apply a negative sign: $C_{21} = -M_{21} = -6$

Element $a_{22} = -1$; **Minor** $\mathbf{M_{22}}$: Remove the second row and second column: $\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = (1 \times 1) - (3 \times 0) = 1$

Cofactor C_{22}: Since $(2 + 2)$ is even, keep the positive sign: $C_{22} = +M_{22} = 1$

Element $a_{23} = 2$; **Minor** $\mathbf{M_{23}}$: Remove the second row and third column: $\begin{vmatrix} 1 & 0 \\ 0 & -2 \end{vmatrix} = (1 \times -2) - (0 \times 0) = -2$

Cofactor C_{23}: Since $(2 + 3)$ is odd, apply a negative sign: $C_{23} = -M_{23} = 2$

Example: Finding the determinant using Minors and Cofactors Method

Step 2: Calculate **the Determinant of A** Using the Second Row

Using the cofactor expansion along the second row:

$$\det(A) = a_{21} \cdot C_{21} + a_{22} \cdot C_{22} + a_{23} \cdot C_{23}$$

Substitute the values:

$$\det(A) = (4 \times -6) + (-1 \times 1) + (2 \times 2) = -24 - 1 + 4 = -21$$

The Inverse of $n \times n$ Matrix

Normally, there are two ways to find the inverse of a n x n matrix.

Either using the *Augmented matrix method* or the *Adjoint method*.

We will only be looking at the *Adjoint method*.

The Inverse of $n \times n$ Matrix

Matrix A: An n × n *square* matrix.

Matrix of Minors (M): The matrix of minors is formed by the determinants of each element's submatrix after removing its row and column.

Matrix of Cofactors (C): The matrix of cofactors is the matrix of minors with each element multiplied by $(-1)^{i+j}$, applying a checkerboard pattern of signs.

Adjoint of A (adj(A)): The transpose of the matrix of cofactors.

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

matrix of cofactors

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}^{t}$$

adjoint matrix

The Inverse of $n \times n$ Matrix

The inverse of A, denoted as A^{-1}, is the adjoint of A divided by the determinant of A.

$$A^{-1} = \frac{1}{\det(A)} \cdot \operatorname{adj}(A), \ \det(A) \neq 0$$

The Inverse of $n \times n$ Matrix

Calculating the Inverse of a *n x n* matrix requires more steps and calculations compared to a 2 x 2 matrix.

The following steps are needed:

Step 1: Make sure that the *determinant of the matrix* is not 0.

Step 2: Calculate the *Matrix of Minors*,

Step 3: Turn that into the *Matrix of Cofactors* by multiplying in the checkerboard manner

Step 4: Transpose; in other words, calculate the *Adjoint matrix*

Step 5: Multiply that by *1/Determinant*.

Example 1:
the Inverse of 3× 3 Matrix

Let's consider a 3x3 **matrix *A***:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$$

Calculate the **Determinant of *A*:** The *determinant of A is* 22, calculated using the standard method for a 3x3 matrix.

Calculate the **Matrix of Minors for *A***: The matrix of minors is found by calculating the determinant for each 2x2 submatrix obtained by removing one row and one column from the original matrix at each element's position:

$$\text{Minors} = \begin{bmatrix} 24 & -5 & -4 \\ 12 & 3 & -2 \\ -2 & 5 & 4 \end{bmatrix}$$

Example 1:
the Inverse of 3× 3 Matrix

Calculate the **Matrix of Cofactors for A:**

The matrix of cofactors is obtained by applying $(-1)^{i+j}$ to each element of the matrix of minors:

$$\text{Cofactors} = \begin{bmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{bmatrix}$$

Calculate the **Adjoint of A:**

The adjoint matrix, denoted as adj(A), is the transpose of the matrix of cofactors:

$$\text{adj}(A) = \begin{bmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{bmatrix}$$

Example 1:
the Inverse of 3× 3 Matrix

Calculate the **Inverse of *A*:**

The inverse of matrix A, denoted as A^{-1}, is found by dividing the adjoint of A by the determinant of A:

$$A^{-1} = \frac{1}{22} \cdot \operatorname{adj}(A) = \begin{bmatrix} 1.09090909 & -0.54545455 & -0.09090909 \\ 0.22727273 & 0.13636364 & -0.22727273 \\ -0.18181818 & 0.09090909 & 0.18181818 \end{bmatrix}$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

In a simple encryption scheme, a message is represented as a vector and encrypted by multiplying it with a key matrix. To decrypt the message, you need the inverse of the key matrix.

Problem Statement:

Suppose you have the following *3x3 key matrix K:*

$$K = \begin{pmatrix} 3 & 4 & 2 \\ 1 & 5 & 7 \\ 6 & 0 & 3 \end{pmatrix}$$

You used this matrix to encrypt a *message represented by the vector* $M = \begin{pmatrix} 12 \\ 5 \\ 10 \end{pmatrix}$, resulting in an *encrypted vector E* given by:

$$E = K \times M$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

Task:

1. Calculate the encrypted message E.

2. Find the inverse of the key matrix K using the determinant, minors, cofactors, and adjoint matrix.

3. Use the inverse of K to decrypt E and recover the original message vector M.

Example 2: Message Encryption and Decryption Using Matrix Inversion

<u>Step 1:</u> Calculate the **Encrypted Message E**

Using matrix multiplication, we calculate $E = K \times M$: $E = \begin{pmatrix} 3 & 4 & 2 \\ 1 & 5 & 7 \\ 6 & 0 & 3 \end{pmatrix} \times \begin{pmatrix} 12 \\ 5 \\ 10 \end{pmatrix}$

Calculate each element of E:

- First row: $(3 \times 12) + (4 \times 5) + (2 \times 10) = 36 + 20 + 20 = 76$
- Second row: $(1 \times 12) + (5 \times 5) + (7 \times 10) = 12 + 25 + 70 = 107$
- Third row: $(6 \times 12) + (0 \times 5) + (3 \times 10) = 72 + 0 + 30 = 102$

So, the encrypted message vector E is: $E = \begin{pmatrix} 76 \\ 107 \\ 102 \end{pmatrix}$

Example 2: Message Encryption and Decryption Using Matrix Inversion

Step 2: Find the **Inverse of Matrix K**

To decrypt the message, we need the inverse of K, denoted K^{-1}.

The initial step is to find the $\mathbf{det}(\boldsymbol{K})$.

Rewrite the first two columns of K to the right of the matrix:

$$\begin{pmatrix} 3 & 4 & 2 & | & 3 & 4 \\ 1 & 5 & 7 & | & 1 & 5 \\ 6 & 0 & 3 & | & 6 & 0 \end{pmatrix}$$

$\det(K) = 3 \times 5 \times 3 + 4 \times 7 \times 6 + 2 \times 1 \times 0 - 6 \times 5 \times 2 - 0 \times 7 \times 3 - 3 \times 1 \times 4 = 213 - 72 = 141$ (not 0, so this matrix is a non-singular one)

Example 2: Message Encryption and Decryption Using Matrix Inversion

Matrix of Minors:

$\mathbf{M_{11}} = 3: \begin{vmatrix} 5 & 7 \\ 0 & 3 \end{vmatrix} = (5 \times 3) - (7 \times 0) = 15;\ \mathbf{M_{12}} = 4: \begin{vmatrix} 1 & 7 \\ 6 & 3 \end{vmatrix} = (1 \times 3) - (7 \times 6) = 3 - 42 = -39$

$\mathbf{M}_{13} = 2: \begin{vmatrix} 1 & 5 \\ 6 & 0 \end{vmatrix} = (1 \times 0) - (5 \times 6) = 0 - 30 = -30;\ \mathbf{M_{21}} = 1: \begin{vmatrix} 4 & 2 \\ 0 & 3 \end{vmatrix} = (4 \times 3) - (2 \times 0) = 12$

$\mathbf{M_{22}} = 5: \begin{vmatrix} 3 & 2 \\ 6 & 3 \end{vmatrix} = (3 \times 3) - (2 \times 6) = 9 - 12 = -3;\ \mathbf{M_{23}} = 7: \begin{vmatrix} 3 & 4 \\ 6 & 0 \end{vmatrix} = (3 \times 0) - (4 \times 6) = 0 - 24 = -24$

$\mathbf{M_{31}} = 6: \begin{vmatrix} 4 & 2 \\ 5 & 7 \end{vmatrix} = (4 \times 7) - (2 \times 5) = 28 - 10 = 18;\ \mathbf{M_{32}} = 0: \begin{vmatrix} 3 & 2 \\ 1 & 7 \end{vmatrix} = (3 \times 7) - (2 \times 1) = 21 - 2 = 19$

$\mathbf{M_{33}} = 3: \begin{vmatrix} 3 & 4 \\ 1 & 5 \end{vmatrix} = (3 \times 5) - (4 \times 1) = 15 - 4 = 11$

$$\text{Minor}(K) = \begin{pmatrix} 15 & -39 & -30 \\ 12 & -3 & -24 \\ 18 & 19 & 11 \end{pmatrix}$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

Matrix of Cofactors:

Apply the checkerboard pattern of signs to the minors:

$$\text{Cofactor}(K) = \begin{pmatrix} 15 & 39 & -30 \\ -12 & -3 & 24 \\ 18 & -19 & 11 \end{pmatrix}$$

Adjoint (Transpose of Cofactor Matrix):

Transpose the cofactor matrix to get the adjoint matrix:

$$\text{Adjoint}(K) = \begin{pmatrix} 15 & -12 & 18 \\ 39 & -3 & -19 \\ -30 & 24 & 11 \end{pmatrix}$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

Calculate the **Inverse**:

The inverse of K is given by:

$$K^{-1} = \frac{1}{\det(K)} \times \text{Adjoint}(K) = \frac{1}{141} \times \begin{pmatrix} 15 & -12 & 18 \\ 39 & -3 & -19 \\ -30 & 24 & 11 \end{pmatrix}$$

Multiply each element by $\frac{1}{141}$:

$$K^{-1} = \begin{pmatrix} \frac{15}{141} & -\frac{12}{141} & \frac{18}{141} \\ \frac{39}{141} & \frac{-3}{141} & \frac{-19}{141} \\ -\frac{30}{141} & \frac{24}{141} & \frac{11}{141} \end{pmatrix}$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

<u>Step 3:</u> Decrypt the Encrypted Message E

To recover the original message vector M, we use:

$$M = K^{-1} \times E$$

Substitute $E = \begin{pmatrix} 76 \\ 107 \\ 102 \end{pmatrix}$ and $K^{-1} = \frac{1}{141}\begin{pmatrix} 15 & -12 & 18 \\ 39 & -3 & -19 \\ -30 & 24 & 11 \end{pmatrix}$:

$$M = \frac{1}{141}\begin{pmatrix} 15 & -12 & 18 \\ 39 & -3 & -19 \\ -30 & 24 & 11 \end{pmatrix} \times \begin{pmatrix} 76 \\ 107 \\ 102 \end{pmatrix}$$

Example 2: Message Encryption and Decryption Using Matrix Inversion

Calculate each element of M by performing the matrix multiplication:

First element of M (top row of K^{-1} with E):

$$M_1 = \frac{1}{141}((15 \times 76) + (-12 \times 107) + (18 \times 102)) = \frac{1}{141}(1140 - 1284 + 1836) = \frac{1}{141} \times 1692 = 12$$

Second element of M (middle row of K^{-1} with E):

$$M_2 = \frac{1}{141}((39 \times 76) + (-3 \times 107) + (-19 \times 102)) = \frac{1}{141}(2964 - 321 - 1938) = \frac{1}{141} \times 705 = 5$$

Third element of M (bottom row of K^{-1} with E):

$$M_3 = \frac{1}{141}((-30 \times 76) + (24 \times 107) + (11 \times 102)) = \frac{1}{141}(-2280 + 2568 + 1122) = \frac{1}{141} \times 1410 = 10$$

Final Answer: The **original message vector M** is: $M = \begin{pmatrix} 12 \\ 5 \\ 10 \end{pmatrix}$

How Matrices are Applied in Computer Science

Computer Graphics: Used for geometric transformations like translation, rotation, and scaling in 2D and 3D graphics.

Machine Learning: Represent and manipulate large datasets; crucial in neural networks for storing weights and activations.

Cryptography: Form the basis of certain encryption algorithms like the Hill Cipher, enhancing security through complex matrix operations.

Network Analysis: Employed in adjacency matrices for representing and analysing network connections in graph theory.

Algorithm Optimisation: Integral in optimising computational efficiency in various algorithms, especially in complex mathematical computations.